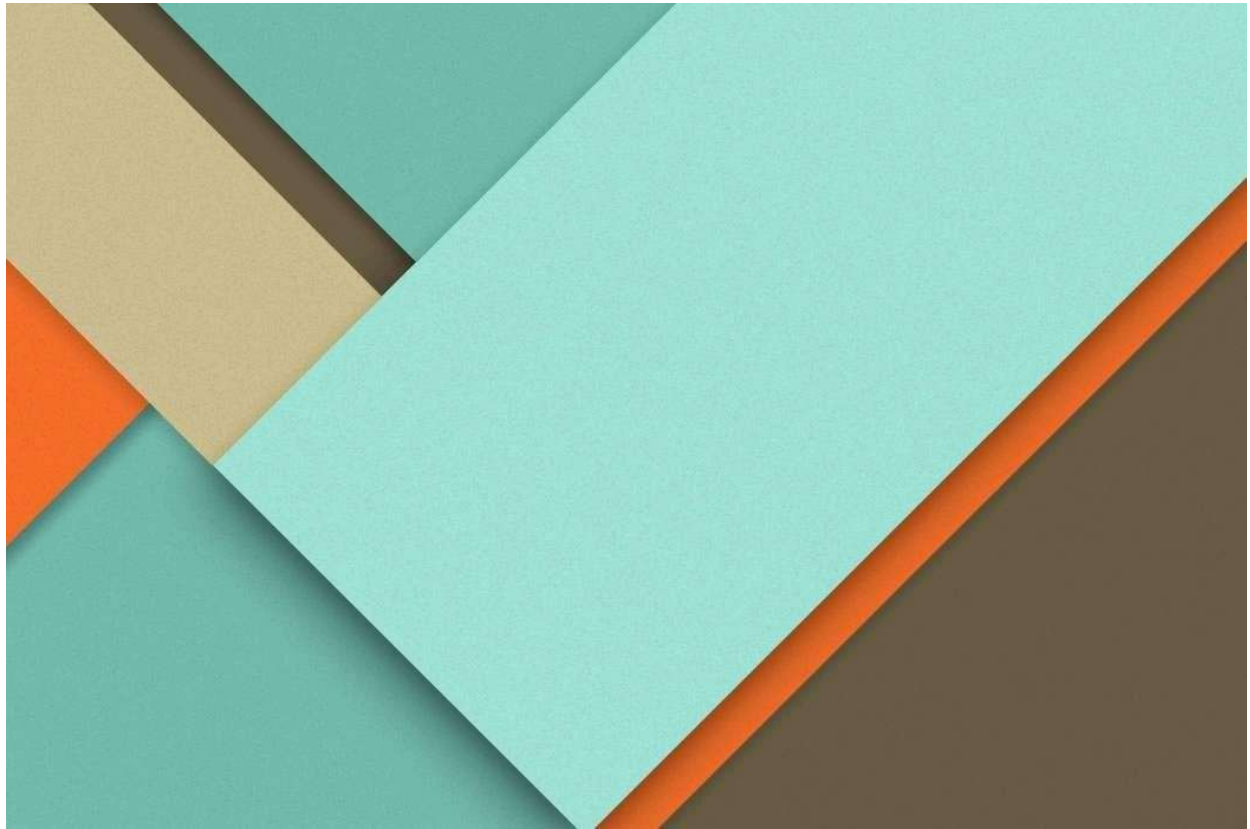




CLOUD COMPUTING

UE23CS351B

LAB 5- Jenkins and Nginx

GITHUB: <https://github.com/akshayag2005/CLOUD-COMPUTING>

DATE: 17/02/2026

NAME: AKSHAY

SRN: PES2UG23CS044

SECTION: A

Task 1:

Set Up Jenkins Using Docker

SS1:

```
[LF]>
[LF]> *****
[LF]> *****
[LF]> *****
[LF]>
[LF]> Jenkins initial setup is required. An admin user has been created and a password generated.
[LF]> Please use the following password to proceed to installation:
[LF]>
[LF]> fbceb7c3814345a1a12ac080be567627
[LF]>
[LF]> This may also be found at: /var/jenkins_home/secrets/initialAdminPassword
[LF]>
[LF]> *****
[LF]> *****
[LF]> *****
```

SS2:

 Jenkins

[+ New Item](#)

[Build History](#)

Build Queue

No builds in the queue.

Build Executor Status

0/2

Welcome to Jenkins!

This page is where your Jenkins jobs will be displayed. To get started, you can set up distributed builds or start building a software project.

Start building your software project

Create a job

Set up a distributed build

Set up an agent

Configure a cloud

Learn more about distributed builds

PE52UG23CS044

ThemeLight

My Views

Account

Appearance

Preferences

Security

Experiments

Credentials

Sign out

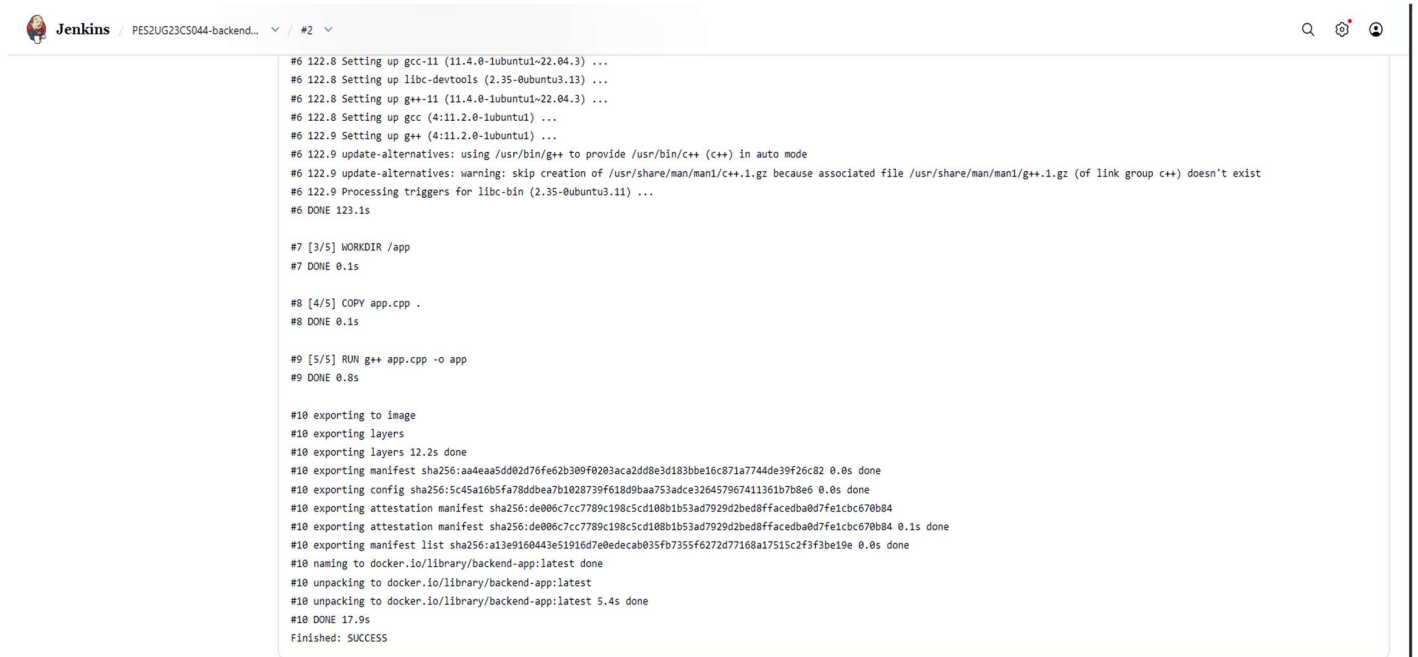
localhost:8080/user/pes2ug23cs044

REST APIJenkins 2.541.1

Task-2:

Create a Jenkins Job to Build the Backend

SS3:



```
Jenkins / PES2UG23CS044-backend... / #2

#6 122.8 Setting up gcc-11 (11.4.0-1ubuntu1~22.04.3) ...
#6 122.8 Setting up libc-devtools (2.35-0ubuntu3.13) ...
#6 122.8 Setting up g++-11 (11.4.0-1ubuntu1~22.04.3) ...
#6 122.8 Setting up gcc (4:11.2.0-1ubuntu1) ...
#6 122.9 Setting up g++ (4:11.2.0-1ubuntu1) ...
#6 122.9 update-alternatives: using /usr/bin/g++ to provide /usr/bin/c++ (c++) in auto mode
#6 122.9 update-alternatives: warning: skip creation of /usr/share/man/man1/g++.1.gz because associated file /usr/share/man/man1/g++.1.gz (of link group c++) doesn't exist
#6 122.9 Processing triggers for libc-bin (2.35-0ubuntu3.11) ...
#6 DONE 123.1s

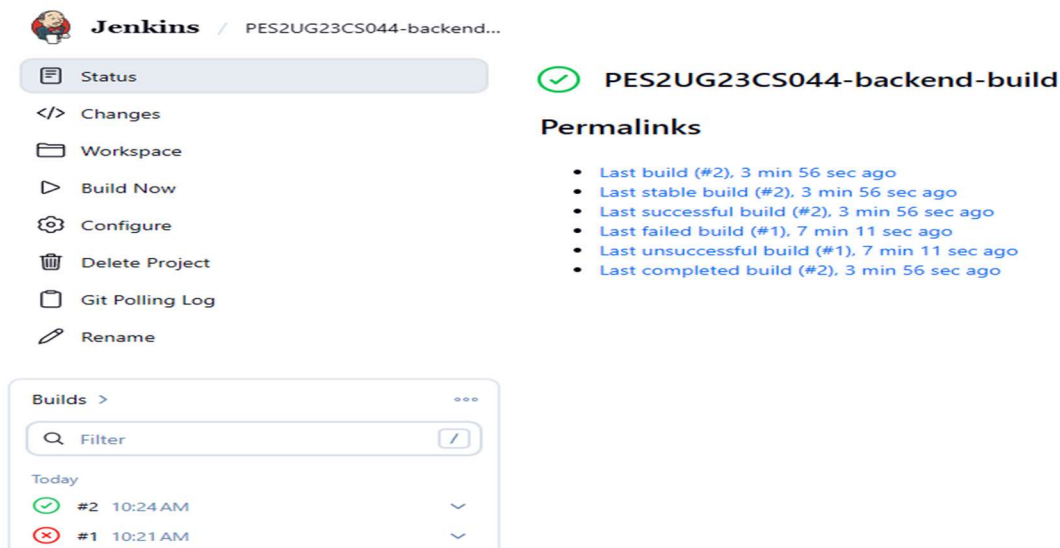
#7 [3/5] WORKDIR /app
#7 DONE 0.1s

#8 [4/5] COPY app.cpp .
#8 DONE 0.1s

#9 [5/5] RUN g++ app.cpp -o app
#9 DONE 0.8s

#10 exporting to image
#10 exporting layers
#10 exporting layers 12.2s done
#10 exporting manifest sha256:aa4eaa5dd02d76fe62b309f0203aca2dd8e3d183bbe16c871a7744de39f26c82 0.0s done
#10 exporting config sha256:5c45a16b5fa78dbba7b1028739f618d9baa753adce326457967411361b7b8e6 0.0s done
#10 exporting attestation manifest sha256:de006c7cc7789c198c5cd108b1b53ad7929d2bed8ffacedba0d7fe1cbc670b84 0.1s done
#10 exporting attestation manifest sha256:de006c7cc7789c198c5cd108b1b53ad7929d2bed8ffacedba0d7fe1cbc670b84 0.1s done
#10 exporting manifest list sha256:a13e9160443e51916d7e0edecab835fb7355f627d77168a17515c2f3f3be19e 0.0s done
#10 naming to docker.io/library/backend-app:latest done
#10 unpacking to docker.io/library/backend-app:latest
#10 unpacking to docker.io/library/backend-app:latest 5.4s done
#10 DONE 17.9s
Finished: SUCCESS
```

SS4:



Jenkins / PES2UG23CS044-backend...

Status

- Changes
- Workspace
- Build Now
- Configure
- Delete Project
- Git Polling Log
- Rename

PES2UG23CS044-backend-build

Permalinks

- Last build (#2), 3 min 56 sec ago
- Last stable build (#2), 3 min 56 sec ago
- Last successful build (#2), 3 min 56 sec ago
- Last failed build (#1), 7 min 11 sec ago
- Last unsuccessful build (#1), 7 min 11 sec ago
- Last completed build (#2), 3 min 56 sec ago

Builds

Filter

Build	Time
 #2	10:24 AM
 #1	10:21 AM

Task-3:

Parameterized Jenkins Job

SS5:

The screenshot shows the Jenkins configuration page for a project named 'Project PES2UG23CS044-backend-build'. The left sidebar contains navigation links: Status, Changes, Workspace, Build with Parameters, Configure, Delete Project, Git Polling Log, and Rename. The main area displays the project name and a message: 'This build requires parameters:'. Below this, a parameter 'BACKEND_COUNT' is defined with the description 'Number of backend containers'. A dropdown menu for this parameter is set to '1'. At the bottom, there are 'Build' and 'Cancel' buttons. A 'Builds' list on the left shows two recent builds: one at 3:41 PM and another at 10:24 AM, both marked as successful.

SS6:

The screenshot displays the Jenkins console output for build #4 of the project 'PES2UG23CS044-backend-build'. The left sidebar shows navigation links: Status, Changes, Console Output, Edit Build Information, Delete build '#4', Parameters, Timings, Git Build Data, and Previous Build. The console output shows the following sequence of commands and their results:

```
Started by user PES2UG23CS044
Running as SYSTEM
Building in workspace /var/jenkins_home/workspace/PES2UG23CS044-backend-build
The recommended git tool is: NONE
No credentials specified
> git rev-parse --resolve-git-dir /var/jenkins_home/workspace/PES2UG23CS044-backend-build/.git # timeout=10
Fetching changes from the remote Git repository
> git config remote.origin.url https://github.com/akshayag2005/PES2UG23CS044_jenkins.git # timeout=10
Fetching upstream changes from https://github.com/akshayag2005/PES2UG23CS044_jenkins.git
> git --version # timeout=10
> git --version # 'git version 2.47.3'
> git fetch --tags --force --progress -- https://github.com/akshayag2005/PES2UG23CS044_jenkins.git +refs/heads/*:refs/remotes/origin/* # timeout=10
> git rev-parse refs/remotes/origin/main^{commit} # timeout=10
Checking out Revision e6ba2726f99ad851fb321889a02b0a81694065e3 (refs/remotes/origin/main)
> git config core.sparsecheckout # timeout=10
> git checkout -f e6ba2726f99ad851fb321889a02b0a81694065e3 # timeout=10
Commit message: "Add files via upload"
> git rev-list --no-walk e6ba2726f99ad851fb321889a02b0a81694065e3 # timeout=10
[PES2UG23CS044-backend-build] $ /bin/sh -xe /tmp/jenkins4597283456681607652.sh
+ docker rm -f backend1 backend2
Error response from daemon: No such container: backend1
Error response from daemon: No such container: backend2
+ [ 1 = 1 ]
+ docker run -d --name backend1 backend-app
88315c2b7b8acfacd478f5de40b781e80df6e5d6365443e74875c192ea64d9a6
Finished: SUCCESS
```

Task-4:

Jenkins Pipeline for Automated Deployment

SS7:

Jenkins / LAB6-PIPELINE-NGINX / #4 / Stages

✓ < #4

Started by PES2UG23CS044 Started 45 sec ago Queued 6 ms Took 13 sec

Graph

Start → Checkout SCM → Build Backend Image → Deploy Backends → Deploy NGINX Load... → End

Deploy NGINX Load Balancer 1s Started 34s ago Jenkins

```
docker rm -f nginx-lb || true
docker run -d --name nginx-lb -p 80:80 -v $(pwd)/CC_LAB-6/nginx/etc/nginx/conf.d nginx
```

0 + docker rm -f nginx-lb
1 nginx-lb
2 + pwd
3 + docker run -d --name nginx-lb -p 80:80 -v /var/jenkins_home/workspace/LAB6-PIPELINE-NGINX/CC_LAB-6/nginx/etc/nginx/conf.d nginx
4 78bd53c264445d68f7ec7327ed779a3afa48b55823b142cdca5f4413cecbae3e

SS8:

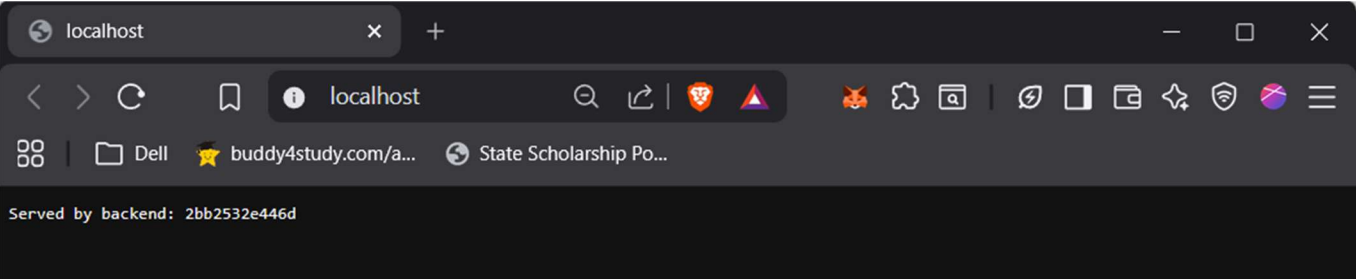
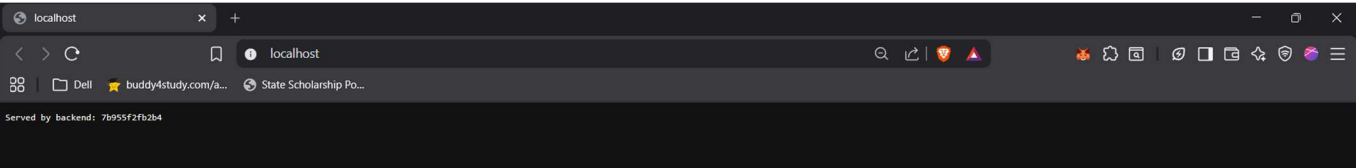
Jenkins / LAB6-PIPELINE-NGINX / #4

```
#10 unpacking to docker.io/library/backend-app:latest 0.0s done
#10 DONE 0.2s
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Deploy Backends)
[Pipeline] sh
+ docker rm -f backend1 backend2
backend1
backend2
+ docker run -d --name backend1 backend-app
b260869591c78604534fcca95fec5b53ea47d397d19675e7115c9cfac122af4d
+ docker run -d --name backend2 backend-app
ed7d94648f3e5e5bb9e2bd98a268e378c3b7b1b7ae5850a5c8348a77ff5e513
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Deploy NGINX Load Balancer)
[Pipeline] sh
+ docker rm -f nginx-lb
nginx-lb
+ pwd
+ docker run -d --name nginx-lb -p 80:80 -v /var/jenkins_home/workspace/LAB6-PIPELINE-NGINX/CC_LAB-6/nginx/etc/nginx/conf.d nginx
78bd53c264445d68f7ec7327ed779a3afa48b55823b142cdca5f4413cecbae3e
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

Jenkins 2.541.1



Browser output:

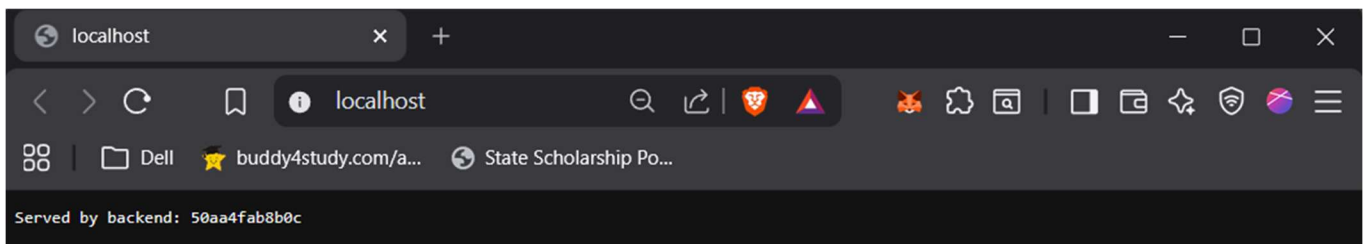
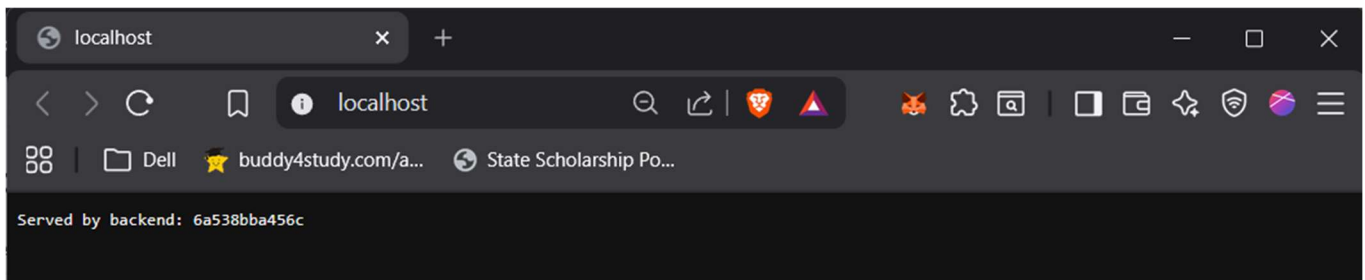


Task 5:

Understanding NGINX Load Balancing Strategies

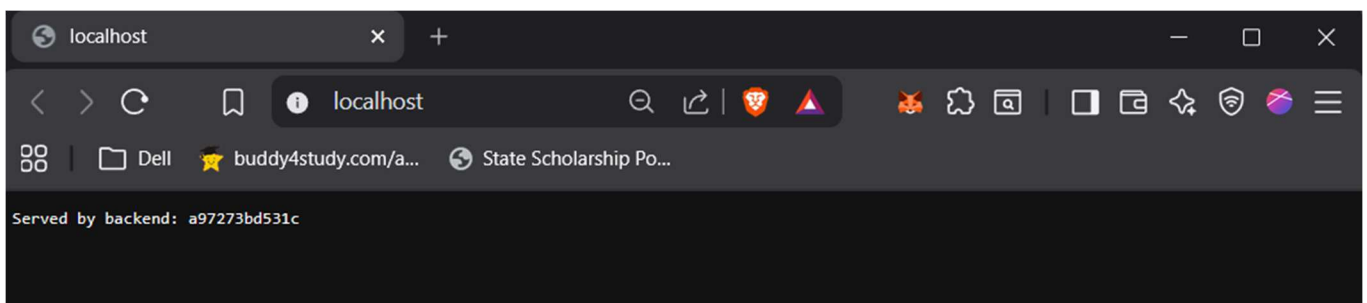
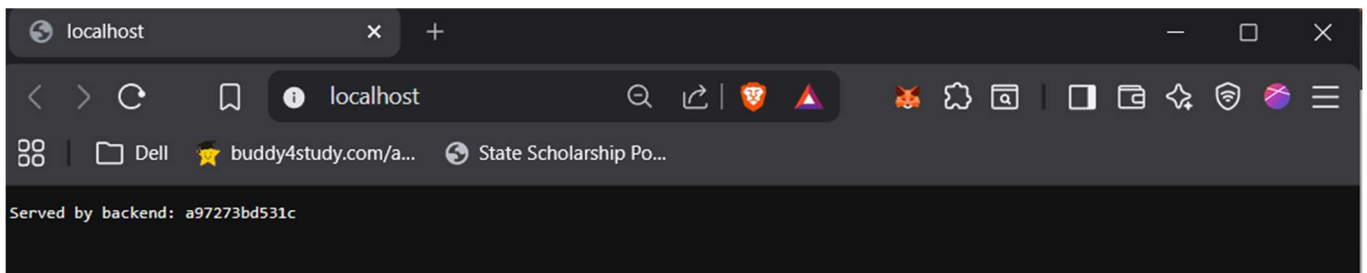
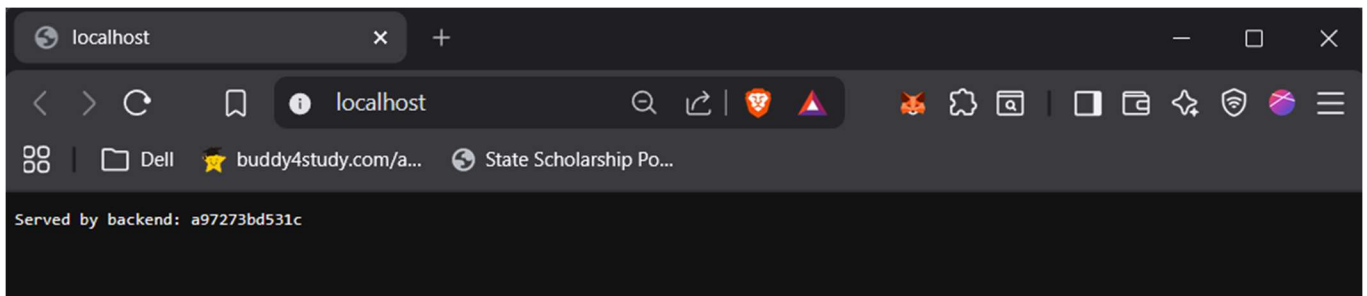
Step 3: Test Least Connections Strategy

Browser output screen shots



Step 4: Test IP Hash Strategy

Browser output screen shots



```
(akshay@AKSHAY)-[/mnt/c/Users/AKSHAY_AG/CC_LAB-6]
$ curl -s http://localhost
curl -s http://localhost
curl -s http://localhost
Served by backend: a97273bd531c
```